Advanced Engineering Maths

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Advanced Engineering Maths (A11943)

Short Title:	Advanced Engineering Maths		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Mathematics and Physics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

To provide an introduction to the theory and techniques of partial differential equations, integral transforms, and the methods needed to develop and implement analytical and numerical solutions to problems arising in electronic engineering applications.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (WD_EAUTO_B)	3 / 5 / M
MTHE-0032	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	3 / 5 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	3 / 5 / M
MTHE-0032	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	3 / 5 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	3 / 5 / M
MTHE-0032	BEng (Hons) (WD_ETRIC_B)	3 / 5 / M
MTHE-0032	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	3 / 5 / M

Indicative Content

- Series solution to ordinary differential equations; special functions: Legendre, Hermite, Bessel etc.; Sturm-Liouville problems: eigenvalue problems and expansions in orthogonal functions (generalised Fourier analysis)
- Fourier Analysis: Fourier transform, DFT, FFT algorithm, convolution and correlation
- Discrete functions, linear and nonlinear difference equations, stability of solutions, mapping from continuous to discrete systems,solution of linear equations using z-transforms
- Partial Differential Equations: classification; wave, diffusion, Poissons and Maxwell's equations; canonical forms; separation of variables over rectangular, cylindrical and spherical domains; transform methods

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Derive, interpret and apply properties of special functions.
2. Construct and interpret solutions of partial differential equations that arise in engineering problems (see content).
3. Demonstrate knowledge of physical contexts in which various partial differential equations arise and the types of boundary conditions needed for well-posedness.
4. Apply the theory covered in this module in terms of progressing from a physical problem, to selecting and implementing an appropriate solution strategy, and ultimately interpreting and representing the results.

Learning and Teaching Methods

- Formal lectures will provide the theoretical base for the subject as well as covering its practical application.
- Tutorial examples, with subsequent discussion and explanation, together with feedback from the in-class tests and suggested reading will support and direct the students in their own personal study. Computer laboratory sessions will be used to demonstrate implementations of methods and provide framework for feedback and assistance on project work.
- Tutorials will focus on solving problems by hand calculation and using Mathlab.

Assessment Methods

	Weighing	**Outcomes Assessed**
Final Written Examination	70%	1,2,3,4
Continuous Assessment	30%	1,2,3,4
Assignment	15%	1,2,3,4
Practical	15%	1,2,3,4

Assessment Criteria

Learning Modes

Learning Mode	**F/T Hours**	**P/T Hours**
Lecture	36	
Seminar	12	
Independent Learning	87	

Supplementary Material

- "Math Works." www.mathworks.com
- DuChateau, P. and D.G. Zachmann. _Applied Partial Differential Equations_. US: Harper &Row, 2002.
- Greenberg, M. _Advanced Engineering Mathematics_. 2nd ed.. .: Prentice Hall, 1998.
- Mathews, J.H. _NUMERICAL METHODS: Using Matlab: (International Edition)_. 4th ed.. .: Pearson Education, 2004.

Requested Resources

- Room Type: Computer Lab